Histogram Processing

- A **histogram** in digital image processing is a graphical representation of the **distribution** of pixel intensity values (gray levels) in an image. It helps in visualizing the frequency of occurrence of different intensity levels across the entire image.
- **Gray Levels**:
  - The **gray levels** represent the different intensities of light in an image. In an 8-bit image, gray levels are typically represented in the range **[0, 255]**, where **0** represents black (the darkest pixel) and **255** represents white (the brightest pixel). This range is denoted as **[0, L-1]**, where **L** is the total number of possible gray levels in the image (for an 8-bit image, **L = 256**).
- **Histogram Representation**:
  - The **horizontal axis** of the histogram represents the **gray levels** or **pixel intensities** (ranging from 0 to 255 in an 8-bit image).

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  - The **vertical axis** represents the **frequency** or **count** of pixels in the image that have a specific intensity (i.e., the number of pixels with a given gray level).
- **Discrete Function**:
  - The histogram is a **discrete function** because the pixel intensities are discrete values, not continuous. The histogram of a digital image with intensity levels in the range [0, L-1] is a discrete function:
    - $\mathbf{h(r_k)=n_k}$
  - where:
    - $\mathbf{r_k}$: The k^{th} gray level or intensity in the range **[0, L-1]**.
    - $\mathbf{n_k}$ is the number of pixels in the image with the gray level $\mathbf{r_k}$.

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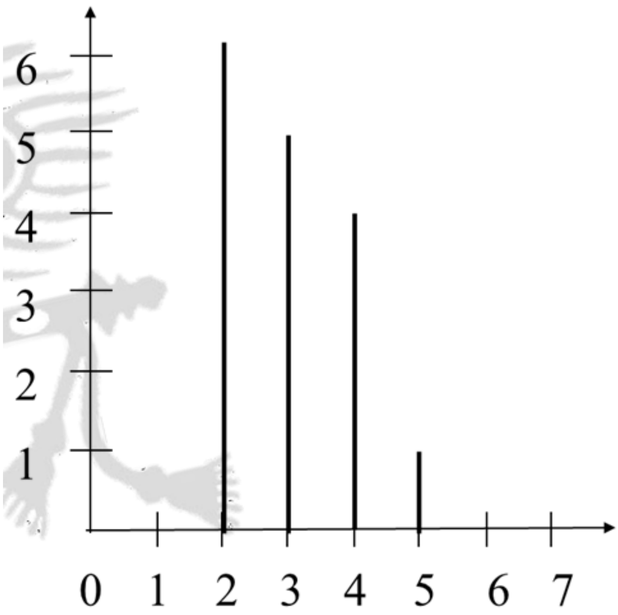
- **h($\mathbf{r_k}$)**: The histogram of a digital image for the gray level $\mathbf{r_k}$

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4



h(2)=6 h(5)=1

h(3)=5 h(6)=0

h(4)=4 h(7)=0

In the histogram plot the X axis represents the gray levels and the Y axis represents the number of pixels in each gray level.

This <u>histogram is normalized</u> by dividing each component by the total number of pixels (n) in the image. Thus, the **normalized histogram** is given by,

$$\mathbf{p(r_k) = n_k/n}$$

– k=0, 1, 2, 3, -----L-1
– $p(r_k)$ gives an estimate of the probability of occurrence of gray level r_k

– The sum of all components of a normalized histogram is equal to 1. i. e. $\sum_{k=0}^{n} P(r_k) = 1$

– Normalized histogram is named as Probability distribution function (PDF).

Q. An image with gray levels between 0 to 7 is given below. Find the histogram of an image and normalize it.

1	6	2	2
1	3	3	3
4	6	4	0
1	6	4	7

Ans: Here, n=total number of pixels=16

r_k	n_k	$P(r_k)=n_k/n$
0	1	1/16
1	3	3/16
2	2	2/16
3	3	3/16
4	3	3/16
5	0	0
6	3	3/16
7	1	1/16

$$\sum_{k=0}^{n} P(r_k) = 1$$

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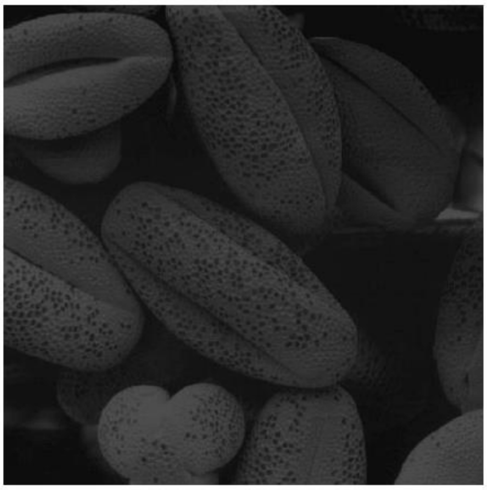
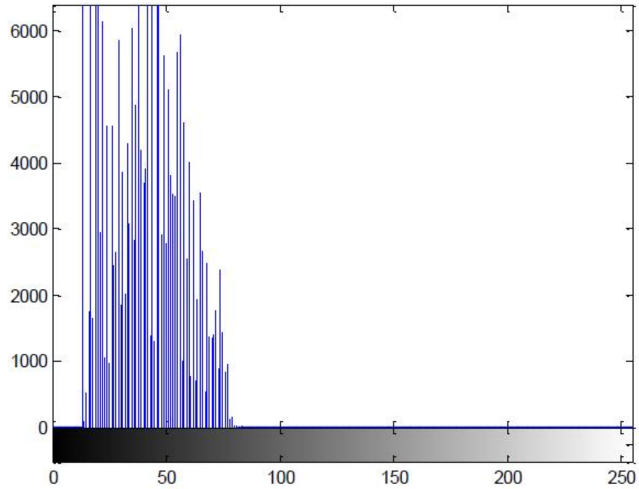
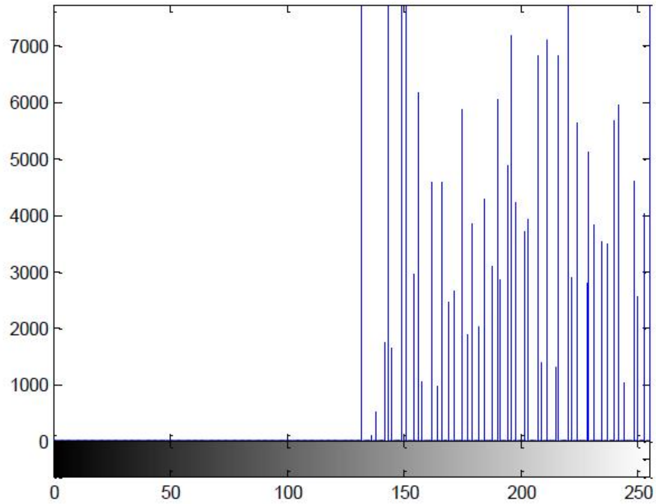
• In the dark images, components of the histogram are concentrated on the lower (dark) side of the gray scale.

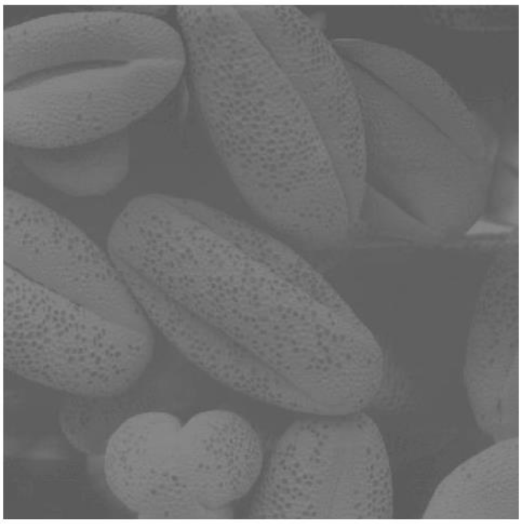


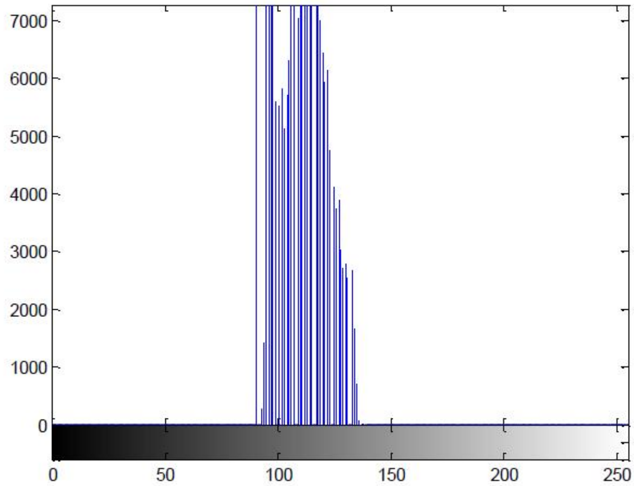


• In bright images, the histogram is biased towards the higher side of the gray.

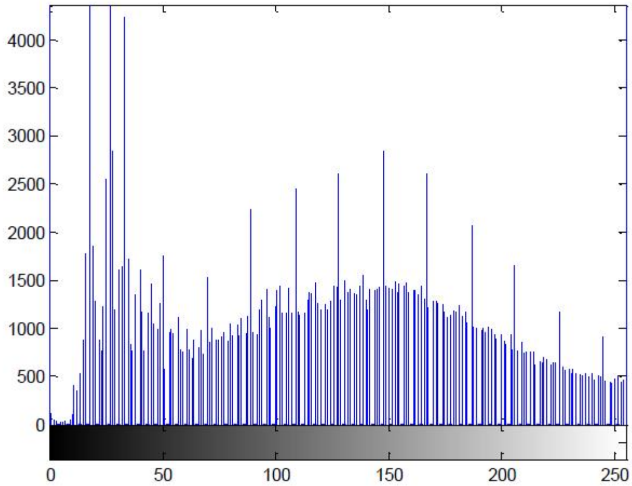


• In low contrast, the histogram will be narrow & centered towards the middle of the grayscale.





- In high contrast, a large variety of gray tones occupy the entire range of possible gray levels.



- **Histogram-based Operations**:
  - **Histogram Equalization**: The process of adjusting the image histogram to enhance its contrast by spreading out the intensity values more uniformly across the available range.

  - **Histogram Matching**: A technique where the image histogram is adjusted to match a predefined or target histogram.

- **Histogram Analysis**:
  - The analysis of the histogram can provide insight into image characteristics such as:
    - **Brightness**: Whether the image is mostly dark or bright.
    - **Contrast**: The range and distribution of pixel intensities.
    - **Dynamic Range**: How widely the pixel values span the entire intensity range.

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Histogram Equalization

A perfect image is one where all the gray levels have equal numbers of pixels.

- **What is it?**
  Histogram equalization is a technique used to adjust the intensity values of an image to make its histogram as uniform (flat) as possible.

- The goal is to enhance the image by evenly distributing the brightness levels across the available intensity range, improving contrast.
- The process modifies the probability distribution of the intensity values (PDF) so that they spread more evenly over the range.
- **Key Advantages**:
  - It enhances contrast automatically, without requiring manual intervention.
  - The results are similar to contrast stretching but offer the benefit of full automation.
- **Main Goal**:

  To find a transformation $s=T(r)$ that produces an image with an approximately flat histogram.
- **Mathematical Conditions for T(r)**:
  - $T(r)$ must be single-valued and monotonically increasing.
  - The transformation must map the intensity range [0,1] to itself, i.e., $0 \le T(r) \le 1$.

The discrete transformation function is given by

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$$s_j = T(\mathrm{r}_j)$$
$$= \sum_{j=0}^{k} p(r_j)$$
$$= \sum_{j=0}^{k} \frac{n_j}{n} \qquad k = 0, 1, \ldots, L-1$$

The following equations bring back the gray levels in the range [0, L-1]

$$s_j = T(\mathrm{r}_j)$$
$$= (\mathrm{L}-1)\sum_{j=0}^{k} p(r_j)$$
$$= (\mathrm{L}-1)\sum_{j=0}^{k} \frac{n_j}{MN}$$

- M – no. of rows
- N – no. of columns
- MN – total no. of pixels in the image

Here S_j is called as Cumulative distribution function (CDF).

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Q. Suppose that a 3-bit image (L = 8) of size 64 × 64 pixels (MN = 4096) has the intensity distribution shown in the following table. Get the histogram equalization transformation function and plot the histogram for each s_k

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Ans:

i/p Gray Level (r_k)	**no. of pixels (n_k)**	**$p(r_k)$ = n_k/MN**	**$\Sigma p(r_k)$**	**s_k =(L-1)Σ**	**o/p Gray Level (s_k) Round off**
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7
		1			

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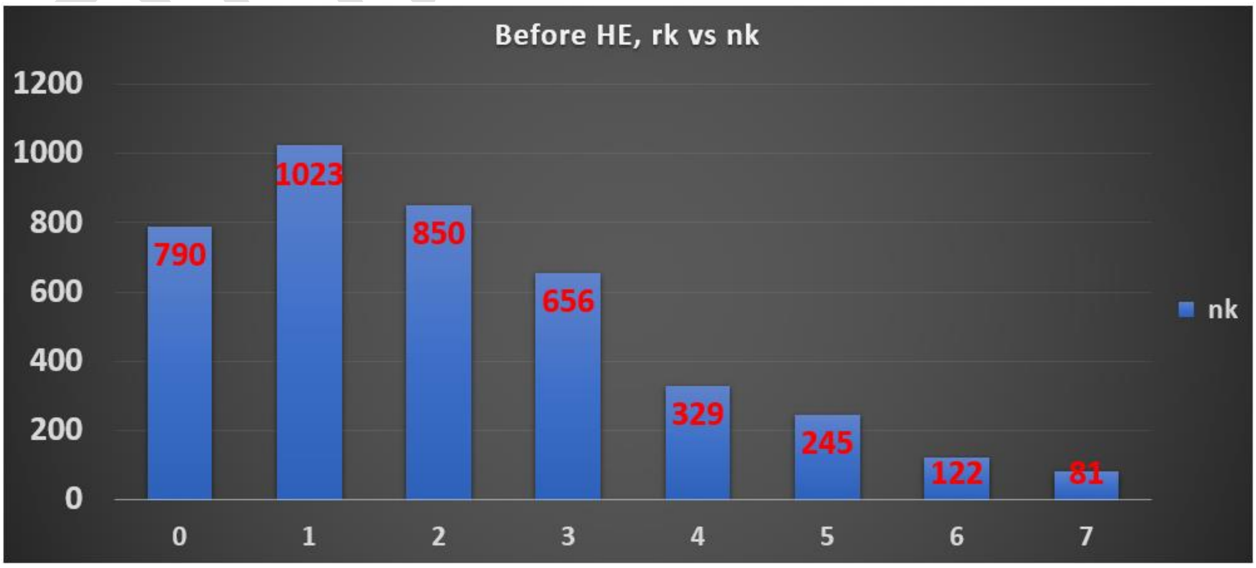
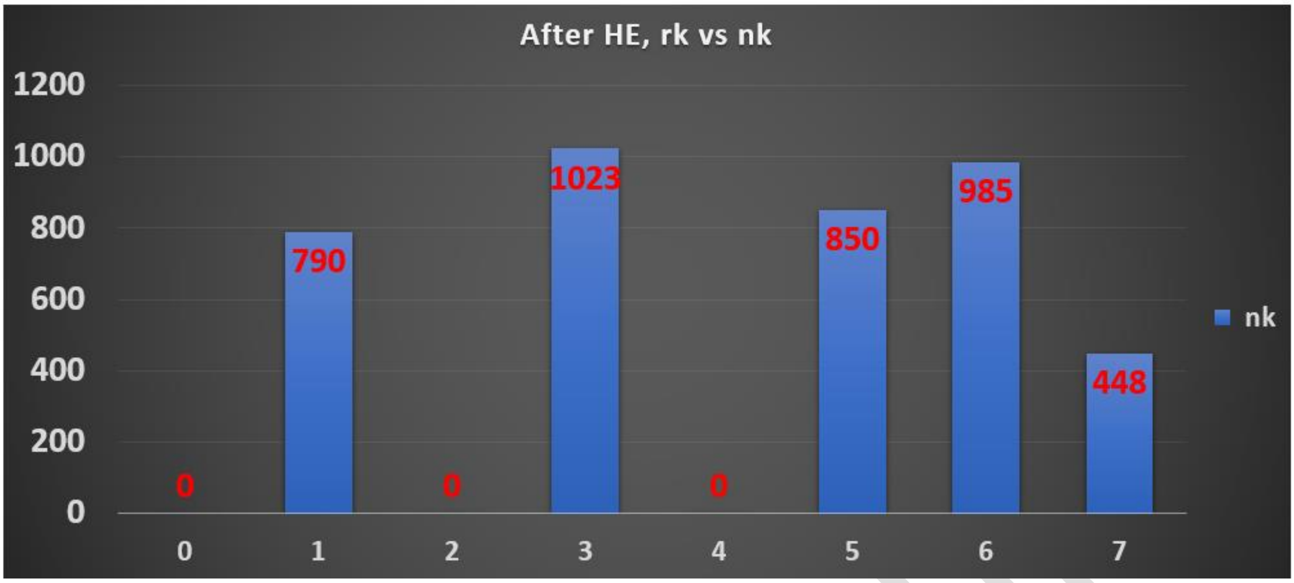
o/p Gray Level (s_k)	no. of pixels (n_k)
1	790
3	1023
5	850
6	656
6	329
7	245
7	122
7	81
	4096

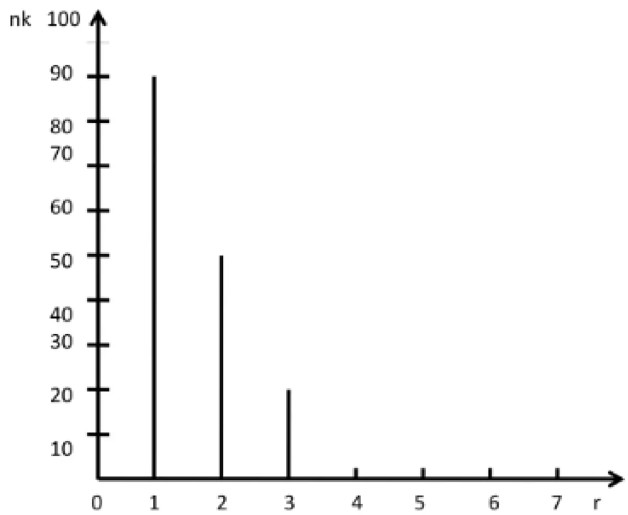
i/p Gray Level (r_k)	no. of pixels (n_k)
0	0
1	790
2	0
3	1023
4	0
5	850
6	985
7	448
	4096





Prob. 2) Equalize the given histogram

Gray Levels (r)	0	1	2	3	4	5	6	7
No. of Pixels	100	90	50	20	0	0	0	0



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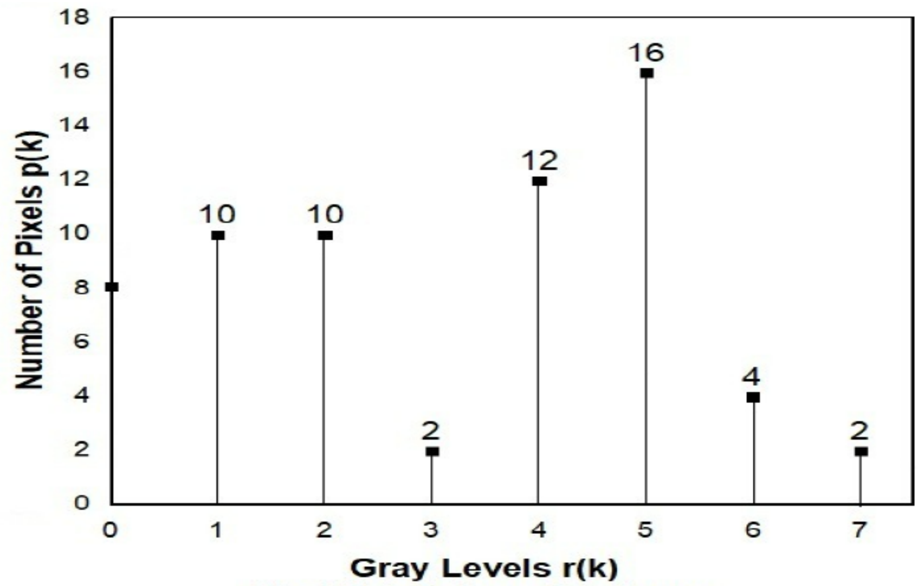
Perform histogram equalization for the 8 X 8, eight level image described in Figure. Also, construct the histogram of the equalized image.	5



Pixel Distribution of an Image

Ans: MN=8x8=64.

i/p Gray Level (r_k)	**no. of pixels (p_k)**	**Pdf= $p(r_k)$ = p_k/MN**	**$\Sigma p(r_k)$**	**s_k =(L-1)Σ**	**o/p Gray Level (s_k) Round off**
0	8	8/64	8/64	0.88	1
1	10	10/64	18/64	1.97	2
2	10	10/64	28/64	3.06	3
3	2	2/64	30/64	3.28	3
4	12	12/64	42/64	4.59	5
5	16	16/64	58/64	6.34	6
6	4	4/64	62/64	6.78	7
7	2	2/64	1.00	7	7
		1			

Calculate the equalized value and draw the graphs for rk vs sk and rk vs new nk.

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rk	New nk
0	0
1	8
2	10
3	12
4	0
5	12
6	16
7	6

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Equalize the following Histogram of a 8 x 8 image. 3

Gray Level	0	1	2	3	4	5	6	7
No. Of pixels	8	10	10	2	12	16	4	2

Ans: Total pixel=64x64=4096. L=8 (total gray level). Equalize the histogram where r_k and n_k values are given.

r_k	**n_k**	**p(r_k) (Pdf)**	**SUM (Cdf)**	**s_k=(L-1)SUM**	**Round off**
0	8	0.125	0.125	0.875	1
1	10	0.156	0.281	1.967	2
2	10	0.156	0.437	3.059	3
3	2	0.0313	0.468	3.276	3
4	12	0.188	0.656	4.592	5
5	16	0.25	0.906	6.342	6
6	4	0.0625	0.969	6.783	7
7	2	0.0313	1	7	7

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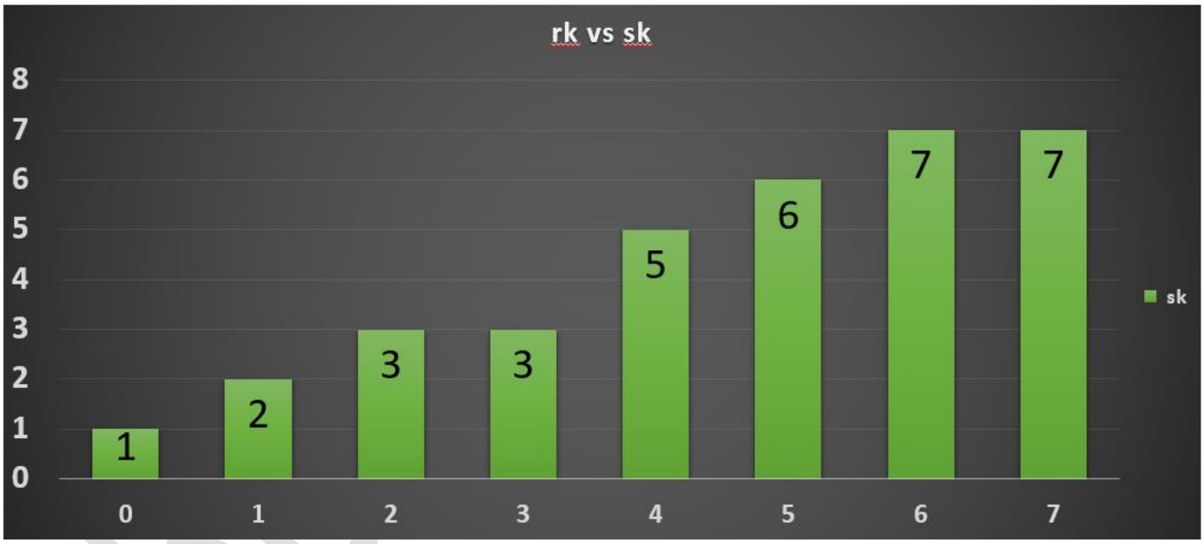
Equalized values of r_k:

s_k	n_k
1	8
2	10
3	10
3	2
5	12
6	16
7	4
7	2

r_k	Equalized n_k
0	0
1	8
2	10
3	12
4	0
5	12
6	16
7	6



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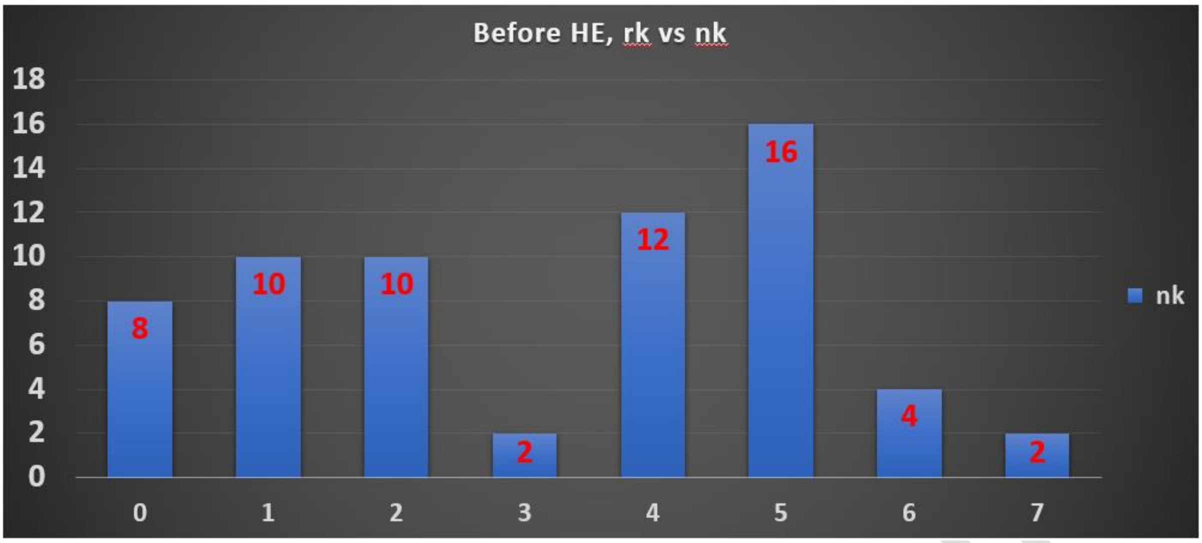
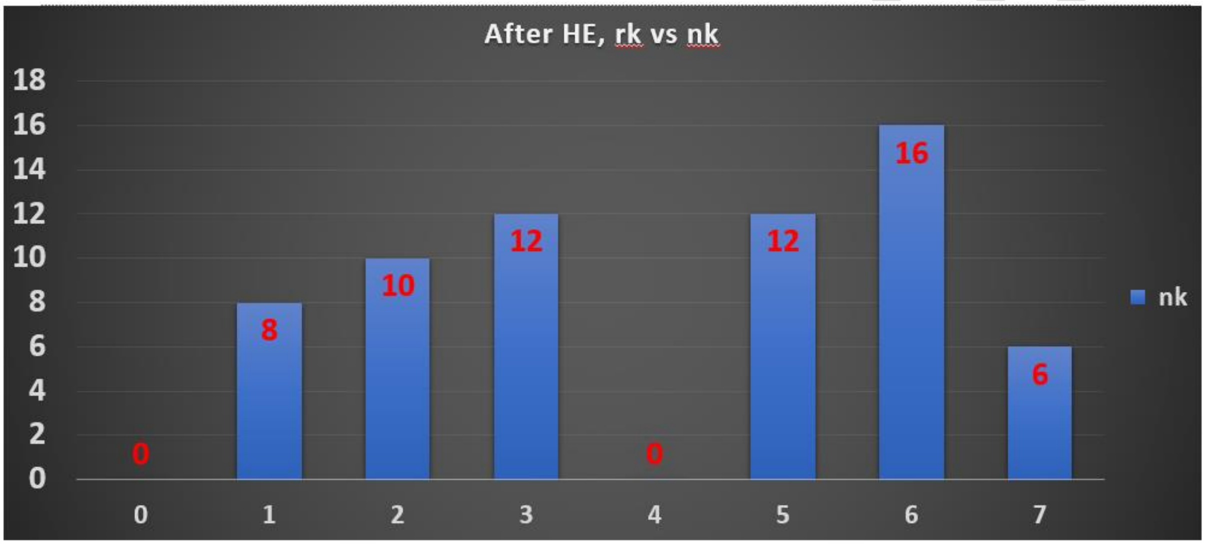
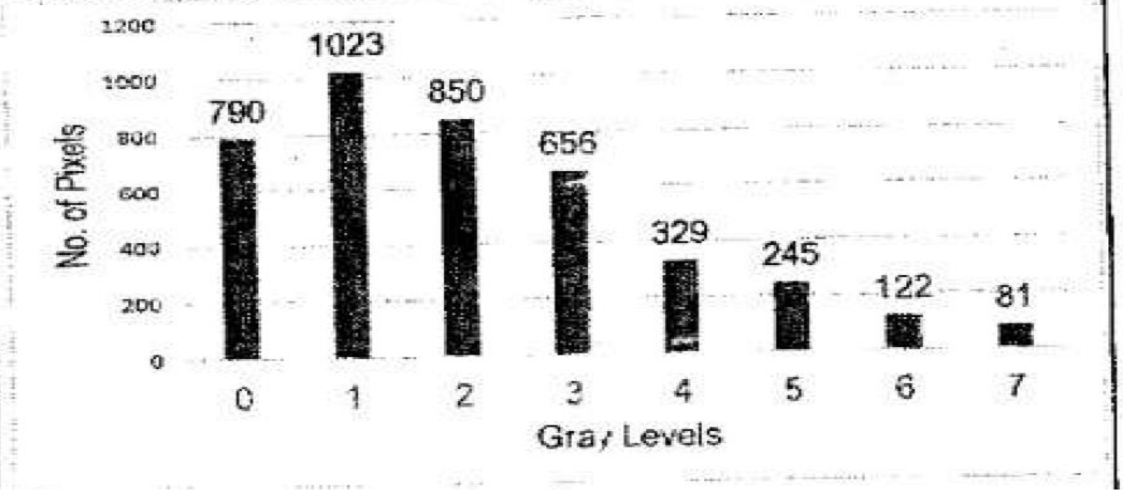




Mid Term 2024 (even)

Following is the Histogram of a 3 bit image. Plot the histogram of the Equalized image. Show all the calculations in tabular format. 5



Solution: For 3-bit, L=2^3=8. In the table match the values from r_k with s_k and write the corresponding n_k (number of pixels).

r_k	**n_k**	**p(r_k)**	**sum**	**(L-1)sum**	**s_k**
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1	7	7

s_k	**n_k**
1	790
3	1023
5	850
6	656
6	329
7	245
7	122
7	81

r_k	**Equalized n_k**
0	0
1	790
2	0
3	1023
4	0
5	850
6	985
7	448

Draw the graphs for r_k vs s_k and r_k vs equalized n_k.

End Term 2024:

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An image I_3 of 9x9 contains a darker object in a lighter background as shown in the Fig 1. 4

11	13	10	11	14	11	13	9	11
8	11	13	12	10	11	10	13	12
10	9	3	5	2	1	4	3	10
8	11	2	4	3	2	0	4	11
11	12	3	6	2	3	3	1	12
9	10	1	3	4	4	5	2	11
13	11	2	2	3	4	3	4	10
12	13	3	4	3	2	5	3	12
11	12	11	9	12	11	11	9	10

Fig. 1

A. Generate the histogram of the image in tabular form.

B. Suggest the gray value threshold to demarcate the object from the background pixels.

C. Find the mean of the object pixels.

Sol: Part A: Generate the histogram of the image in tabular form

To create the histogram:

1. Count the frequency of each gray value.
2. Tabulate the result.

The pixel values in the 9×9 matrix are given.

Now, count the occurrences of each gray value:

To determine the occurrences of each gray value from the given table, I'll systematically count all values from **0 to 14** in the matrix.

Gray Value Frequency or no. of each pixel Count

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Gray Value	Frequency
0	1
1	3
2	8
3	12
4	8
5	3
6	1
8	2
9	5
10	8
11	15
12	8
13	6
14	1

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Part B: Suggest the gray value threshold to demarcate the object from the background pixels

Observations:

1. **Object Pixels**: The darker object is represented by lower gray values (mainly **0, 1, 2, 3, 4, 5, and 6**).

2. **Background Pixels**: The background has higher gray values, primarily **8–14**.

Suggested Threshold:

- A reasonable threshold to separate the object from the background would be **6**.
- Pixels with gray values ≤ 6 belong to the **object**, while values > 6 belong to the **background**.

Part C: Find the mean of the object pixels

To calculate the mean:

1. Identify all pixels that satisfy the condition **gray value ≤ 6** (from Part B).
2. Sum these values and divide by their count.

Object Pixels (gray values ≤ 6):

3, 5, 2, 1, 4, 3, 2, 4, 3, 2, 0, 4, 3, 6, 2, 3, 3, 1, 1, 3, 4, 4, 5, 2, 2, 2, 3, 4, 3, 4, 3, 4, 3, 2, 5, 3.

- **Total Count** = 36
- **Sum of Values** =

  3+5+2+1+4+3+2+4+3+2+0+4+3+6+2+3+3+1+1+3+4+4 +5+2+2+2+3+4+3+4+3+4+3+2+5+3 = 108

Mean:

$$\frac{\text{Sum of Object Pixels}}{\text{Total Count}} = \frac{108}{36} = 3$$

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Limitation of Histogram Equalization: Histogram equalization may not always produce desirable results, particularly if the given histogram is very narrow. It can produce false edges and regions. It can also increase image "graininess" and "patchiness." To mitigate these problems Histogram matching was proposed.

Histogram Matching or Specification

1. Histogram matching, also known as histogram specification, is an advanced enhancement technique used to modify the histogram of an input image to match a given target histogram.
   - Unlike histogram equalization, which aims for a uniform histogram, histogram specification allows for greater flexibility by specifying the desired shape of the histogram.
2. **Objective**
   - The goal is to adjust the intensities of an image such that its histogram resembles a predefined histogram, G(zk).
3. **Steps in Histogram Matching**
   - **Step 1: Compute the Input Histogram**
     Calculate the pdf or normalized histogram of the original image, $p(r_k)$, where r_k represents the intensity levels.
   - **Step 2: Perform Histogram Equalization on the Input Image**
     Transform the input histogram into a cumulative

distribution function (CDF), $S(r_k)$, using the relation:

$$S(r_k) = (L-1)\sum_{j=0}^{k} p(r_j)$$

- **Step 3: Compute the Target image CDF**
  Determine the CDF of the given target histogram, $G(z_k)$:

$$G(z_k) = (L-1)\sum_{j=0}^{k} p(z_j)$$

- **Step 4: Match the CDFs**
  Match each intensity level of the input image to the closest intensity level in the target image based on their CDF values.
- **Step 5: Transform the Input Image**
  Replace the input image's pixel intensities with the corresponding matched intensities to achieve the desired histogram.

4. **Advantages of Histogram Specification**
   - Allows more control over the output image's appearance compared to histogram equalization.
   - Useful in applications where a specific intensity distribution is desired, such as medical imaging or pattern recognition.
   - Helps in standardizing images for comparison or further processing.
5. **Differences from Histogram Equalization**

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- Histogram equalization produces a uniform distribution, while histogram specification can match any desired distribution.
- Histogram specification requires a predefined histogram as input.

6. **Applications**
   - **Medical Imaging**: Enhancing X-rays or CT scans to emphasize specific intensity ranges.
   - **Remote Sensing**: Standardizing satellite images to match a reference histogram.
   - **Forensics**: Improving low-contrast images for better visibility and analysis.
7. **Limitations**
   - Requires a target histogram, which may not always be available.
   - Computationally intensive due to the matching process.

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Q. Suppose that a 3-bit image of size 64×64 pixels has the intensity distribution shown in the table (on the left). Get the histogram transformation function and make the output image with the specified histogram, listed in the 2nd table.

Sol: Table for the actual image:

(rk)	(nk)	p(rk)	Σ	(L-1)Σ	(sk)
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7

Table for the target (specified histogram) image:

(zk)	p(zk)	Σ	(L-1)Σ	G(zk)
0	0.00	0	0	0
1	0.00	0	0	0
2	0.00	0	0	0
3	0.15	0.15	1.05	1
4	0.20	0.35	2.45	2
5	0.30	0.65	4.55	5
6	0.20	0.85	5.95	6
7	0.15	1	7	7

Matching:

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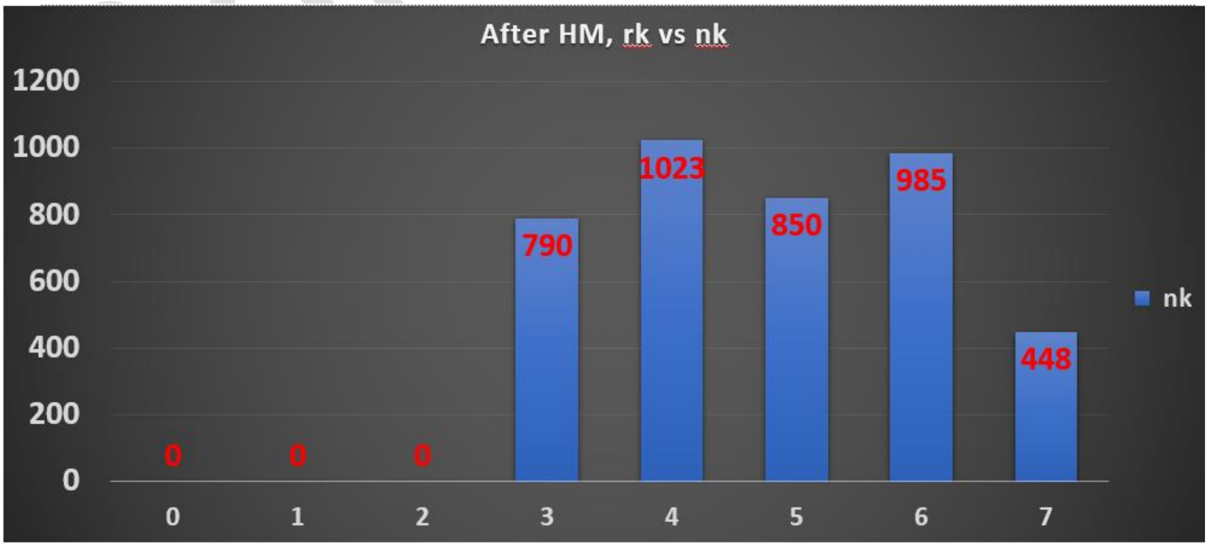
r_k	s_k	G(z_k)	(zk)	Mapped value	n_k
0	1	0	0	**3**	790
1	3	0	1	**4**	1023
2	5	0	2	**5**	850
3	6	1	3	**6**	656
4	6	2	4	**6**	329
5	7	5	5	**7**	245
6	7	6	6	**7**	122
7	7	7	7	**7**	81

r_k	n_k
0	0
1	0
2	0
3	790
4	1023
5	850
6	985
7	448



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Perform the histogram specification on the 8 x 8, eight gray level image histogram described below: 5

r_k	0	1	2	3	4	5	6	7
n_k	2	3	5	6	9	12	14	13

Target Image Histogram:

r_k	0	1	2	3	4	5	6	7
n_k	13	12	14	14	11	0	0	0

Sol: Step 1: Calculate the CDF (Cumulative Distribution Function) or Histogram Equalization for both the given image and the target image histograms.

Given Image Histogram:

r_k = [0, 1, 2, 3, 4, 5, 6, 7]

n_k = [2, 3, 5, 6, 9, 12, 14, 13]

The total number of pixels is 64 (since 8x8). The total no. of gray level (L) is 8.

The CDF for the given image histogram:

r_k	n_k	p(r_k)	SUM	s_k=(L-1)*SUM	s_k Round off
0	2	0.031	0.031	0.217	0
1	3	0.047	0.078	0.546	1
2	5	0.078	0.156	1.092	1
3	6	0.094	0.25	1.75	2
4	9	0.141	0.391	2.737	3
5	12	0.188	0.579	4.053	4
6	14	0.219	0.798	5.586	6
7	13	0.203	1	7	7

Target Image Histogram:

r_k = [0, 1, 2, 3, 4, 5, 6, 7]

n_k = [13, 12, 14, 14, 11, 0, 0, 0]

The CDF for the target image is s_K_T:

r_k	n_k	p(r_k)	SUM	(L-1)SUM	s_k_T
0	13	0.203	0.203	1.421	1
1	12	0.188	0.391	2.737	3
2	14	0.219	0.61	4.27	4
3	14	0.219	0.829	5.803	6
4	11	0.172	1	7	7
5	0	0	1	7	7
6	0	0	1	7	7
7	0	0	1	7	7

Step 2: Map the input CDF values to the target CDF values and take the corresponding value in s_k.

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r_k	s_k	s_k_T	r_k	Mapped value	n_k for mapped value
0	0	1	0	0	2
1	1	3	1	0	3
2	1	4	2	0	5
3	2	6	3	1	6
4	3	7	4	1	9
5	4	7	5	2	12
6	6	7	6	3	14
7	7	7	7	4	13

r_k	New n_k
0	2+3+5=10
1	6+9=15
2	12
3	14
4	13
5	0
6	0
7	0

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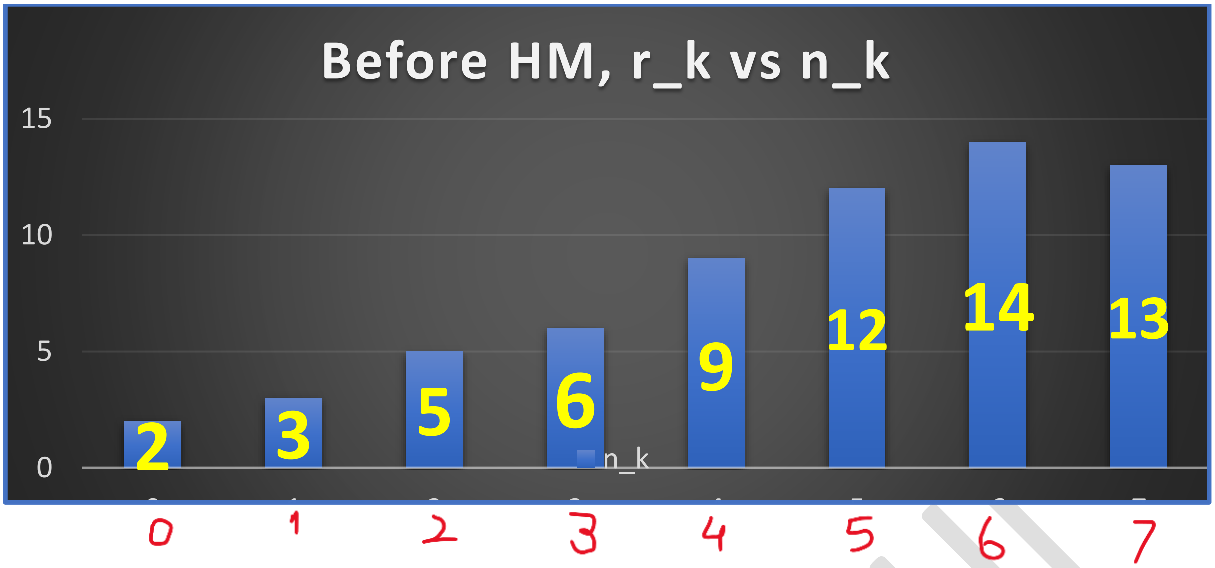

Fig1: Graphical representation of r_k (Given image, 1st table) vs n_k

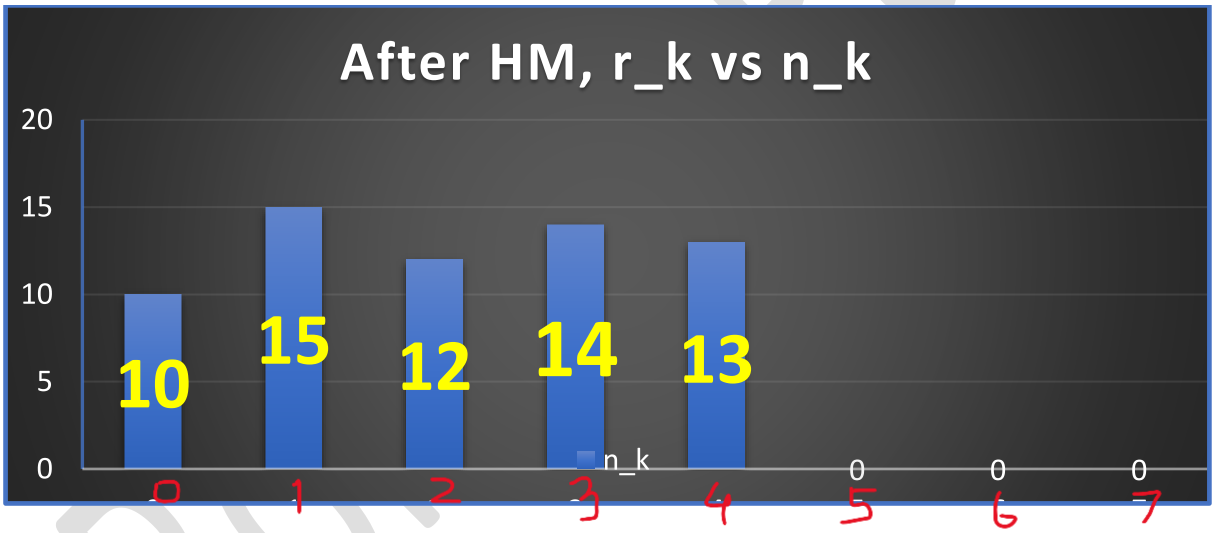

Fig2: Graphical representation of r_k vs New n_k

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Mid-Term 2023-2024 (Odd)

A and B are two 3 – bit images. Histogram Equalization was done on both the images and following results were obtained.

I/P Gray Level of Image A	0	1	2	3	4	5	6	7
O/P Gray Level of Image A	1	3	5	6	6	7	7	7

I/P Gray Level of Image B	0	1	2	3	4	5	6	7
O/P Gray Level of Image B	0	0	0	1	2	5	6	7

3

We want to match the histogram of image A to that of image B. What will be the result of Histogram matching? Give your answer in the following format.

I/P Gray Level of Image A	0	1	2	3	4	5	6	7
O/P Gray Level of Image A after matching								

Solution: (a) Histogram of A and B image

Input gray level of A (r_k)	Output gray level of A (s_k)	Input gray level of B (z_k)	Output gray level of B (G_k)
0	1	0	0
1	3	1	0
2	5	2	0
3	6	3	1
4	6	4	2
5	7	5	5
6	7	6	6
7	7	7	7

Map the values between s_k and G_k i.e. find the values of s_k in G_k. It should be **exactly matched or the nearest value** and write the corresponding input gray value z_k.

Input gray level of A (r_k)	Output gray level of A (s_k)	Output gray level of B (G_k)	Input gray level of B (z_k)	Mapped value or output gray level after HM
0	1	0	0	3
1	3	0	1	4
2	5	0	2	5
3	6	1	3	6
4	6	2	4	6
5	7	5	5	7
6	7	6	6	7
7	7	7	7	7

*Here, 1 is mapped with 1, and the corresponding value in z_k is 3. Similarly, 3 is mapped with 2, and the corresponding value in z_k is 4.

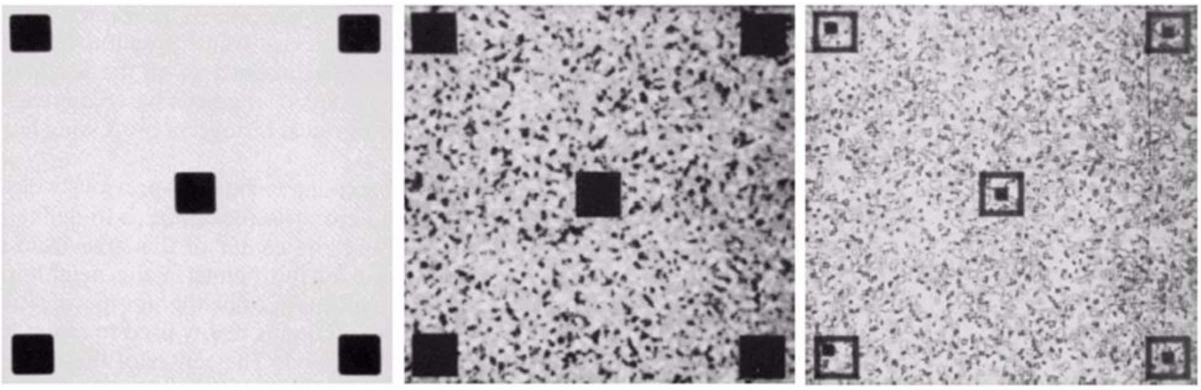
Local Enhancement:

- **Local Enhancement** improves image details in specific areas rather than globally processing the entire image.
- Unlike global methods (e.g. Histogram processing), it uses a transformation function based on the gray levels of a localized region or neighborhood.
- It is ideal for highlighting fine details, handling varying illumination, or enhancing features in small regions.

Bondita Paul

- Examples include adaptive histogram equalization and region-based intensity adjustments.

Local enhancement is essential in scenarios like medical imaging, where subtle features need to be emphasized in localized regions for better analysis.



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